

Why Design For Manufacturing Is India's Electronics Game-Changer

India's electronics growth story will not be written by assembly alone. Embedding Design for Manufacturing into every stage of product creation is the step that can transform ambition into global competitiveness—and make India a genuine leader in electronics.



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India's push toward self-reliance in electronics is being tested across high-stakes sectors—defence, railways, energy, aerospace, and industrial systems—all of which demand hardware that is rugged, reliable, and scalable. The old model of import dependence and assembly is giving way to something far more ambitious: electronic design for manufacturing (DFM). No longer a peripheral practice, DFM has become the foundation for producing hardware that meets global standards while withstanding real-world operating challenges, making it indispensable to India's rise as a serious electronics contender.

The Indian electronics ecosystem: Context and imperatives

Defence, aerospace, railways, oil and gas, and renewable energy are all moving rapidly towards digitisation, creating a

surge in demand for hardware that can withstand harsh conditions while delivering consistent performance. Meeting this demand requires more than incremental upgrades—it calls for a design philosophy that anticipates scale, reliability, and manufacturability from the outset.

For example, Indian Railways is undertaking the large-scale deployment of connected systems to enhance both operational efficiency and safety across its vast network. At the same time, Industrial IoT is transforming key sectors such as oil and gas, manufacturing, and energy, necessitating the use of robust gateways and intelligent edge devices to ensure reliable data processing and connectivity. Meanwhile, the consumer electronics market continues its rapid expansion, fuelling demand for innovative, high-quality products that meet evolving user expectations.